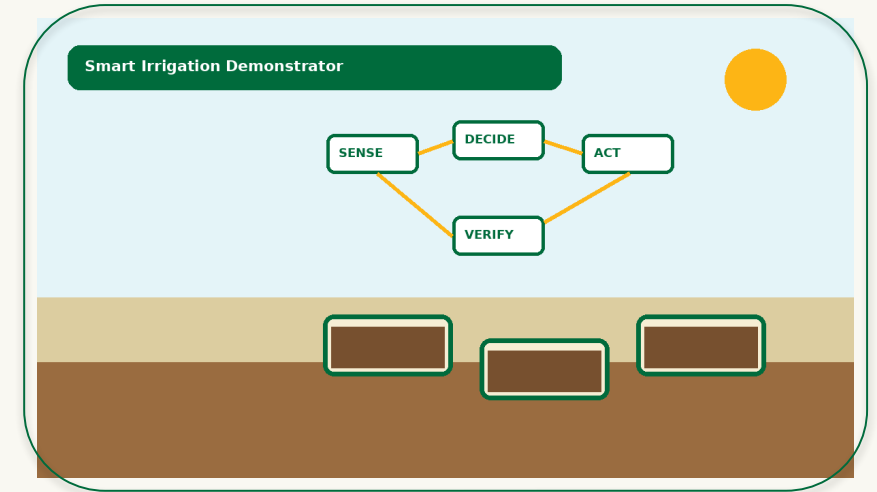


Smart Irrigation Demonstrator

Sensor-Informed Irrigation Control

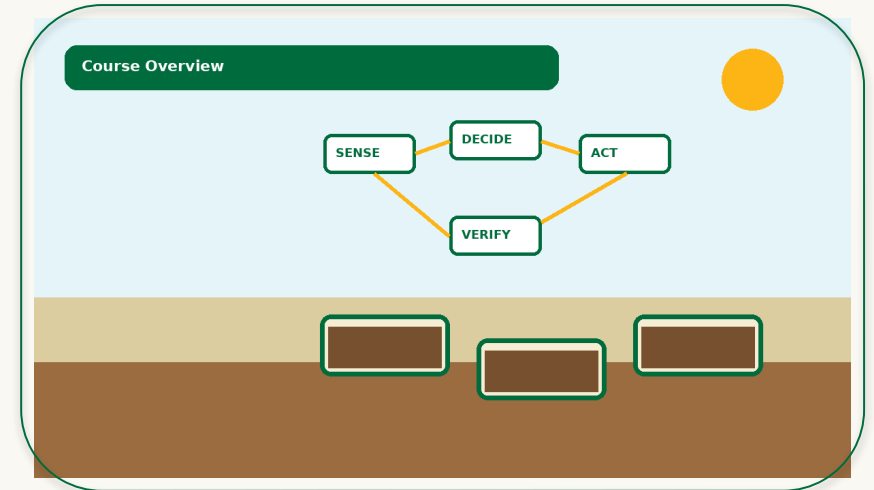
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide quantitative analysis; distinguish measurement from inference; connect evidence to management, engineering and research decisions.
- Core materials: teacher guide, system drawings, approved tools, low-voltage educational hardware where relevant, measuring instruments, meters/sensors, data sheets/logger, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit an engineering design, monitoring plan, experimental protocol or systems-analysis report appropriate to the course.



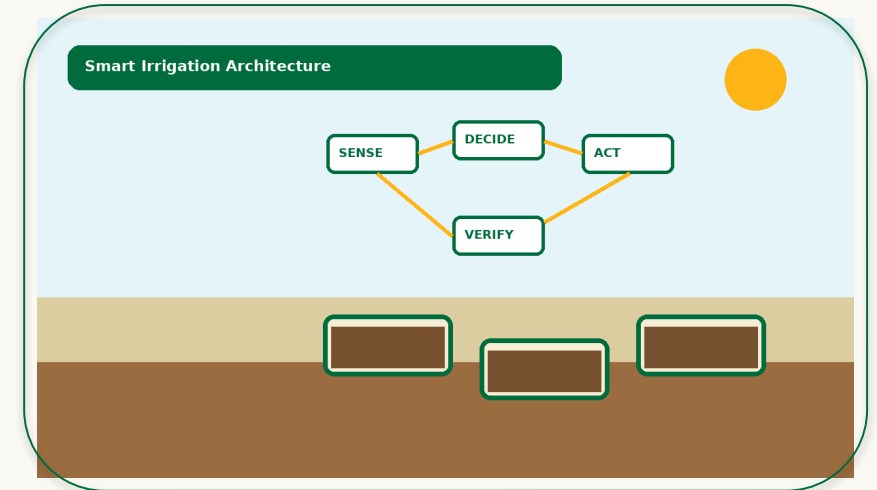
TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

Smart Irrigation Architecture

Teacher technical knowledge, laboratory/field practice and student supervision

- Map sensors, data input, controller/decision logic, valves/actuators, irrigation network and feedback measurements.
- Distinguish sensing, decision and actuation layers.
- Define manual override and safe-failure behavior.
- Use low-voltage, enclosed, finger-safe educational hardware.
- Keep mains power and pump servicing outside student access.



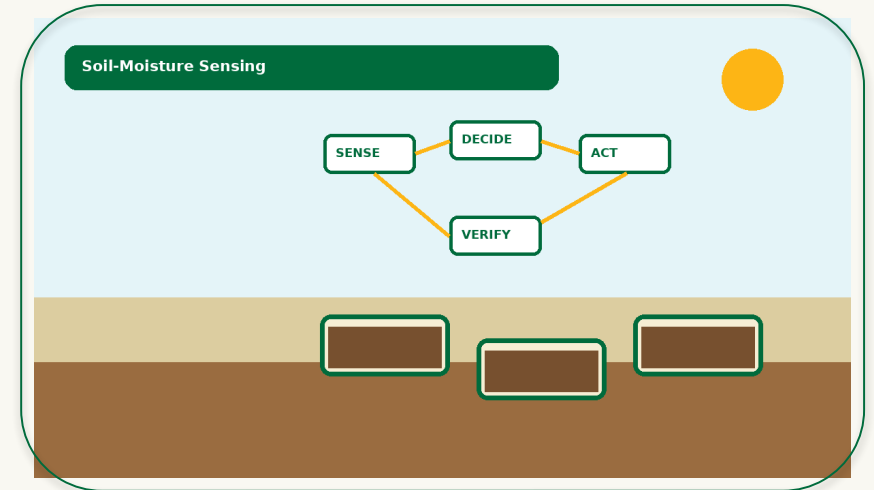
TEACHER PRACTICE

- Teacher task: draw the complete sense-decide-act-verify architecture.

Soil-Moisture Sensing

Teacher technical knowledge, laboratory/field practice and student supervision

- Select representative root-zone sensor locations and depths.
- Understand the sensor output and units used by the installed system.
- Use multiple measurements or zones where soil variability is important.
- Validate sensor readings against a reference/manual check.
- Avoid placing every sensor immediately beside emitters unless intentional.



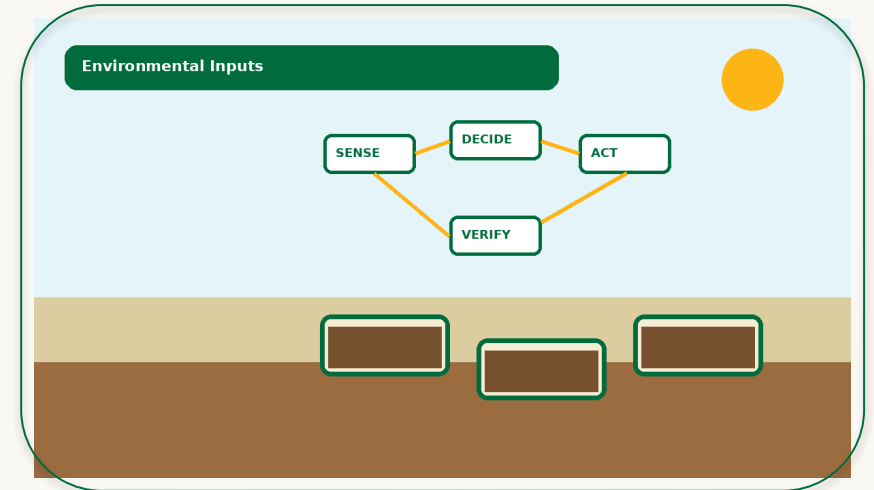
TEACHER PRACTICE

- Prepare a soil-moisture sensor network for two irrigation zones.

Environmental Inputs

Teacher technical knowledge, laboratory/field practice and student supervision

- Use relevant inputs such as rainfall, air temperature, humidity or other available measurements to contextualize irrigation decisions.
- Distinguish measured environmental inputs from forecasts or assumed values.
- Define how each input influences the control rule, if at all.
- Check data quality before using an input for automated action.
- Avoid unnecessarily complex logic that students cannot verify.



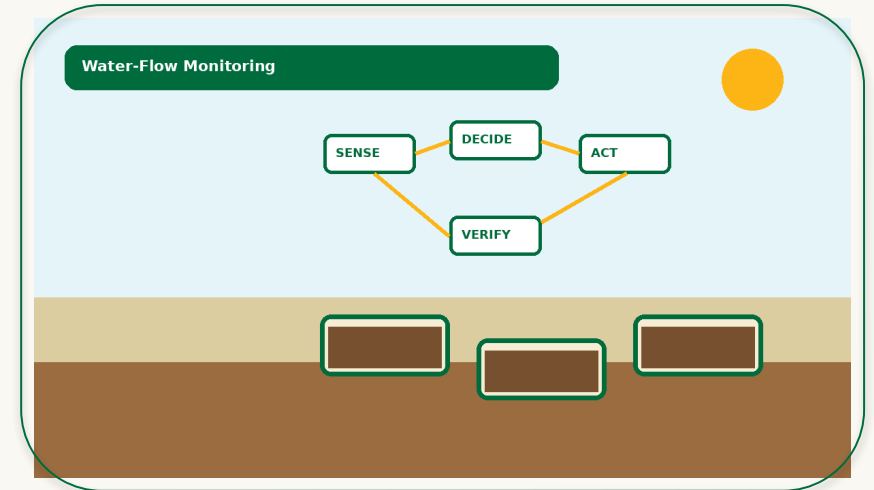
TEACHER PRACTICE

- Create an input-to-decision table.

Water-Flow Monitoring

Teacher technical knowledge, laboratory/field practice and student supervision

- Measure zone flow during irrigation and total delivered volume where instrumentation permits.
- Use flow data to confirm that commanded irrigation actually occurred.
- Detect possible leaks, blockage or valve failure from abnormal readings.
- Record flow with valve/zone status and time.
- Verify flow sensors periodically.



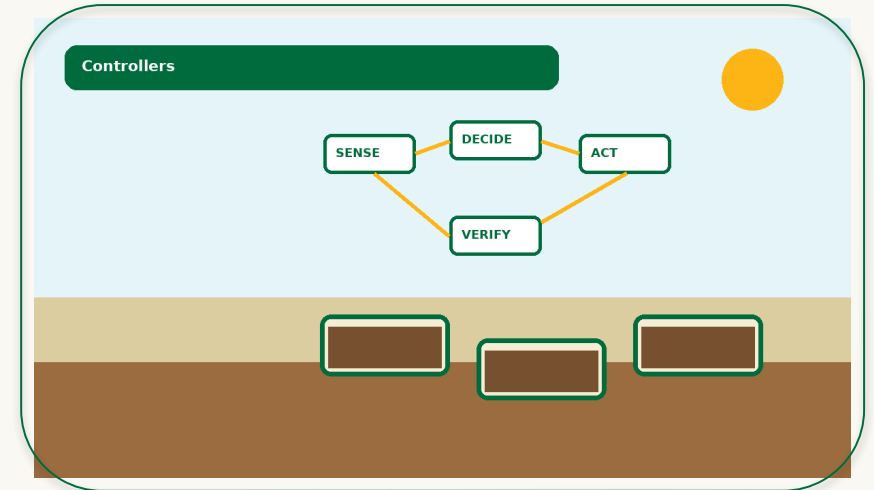
TEACHER PRACTICE

- Develop a flow-based irrigation verification routine.

Controllers

Teacher technical knowledge, laboratory/field practice and student supervision

- Explain the controller as the device executing programmed decision logic.
- Use transparent, teacher-readable rules rather than inaccessible black-box behavior for school demonstrations.
- Document input variables, decision interval and output commands.
- Maintain manual override and emergency stop procedures.
- Students may program approved low-voltage logic but not service mains equipment.



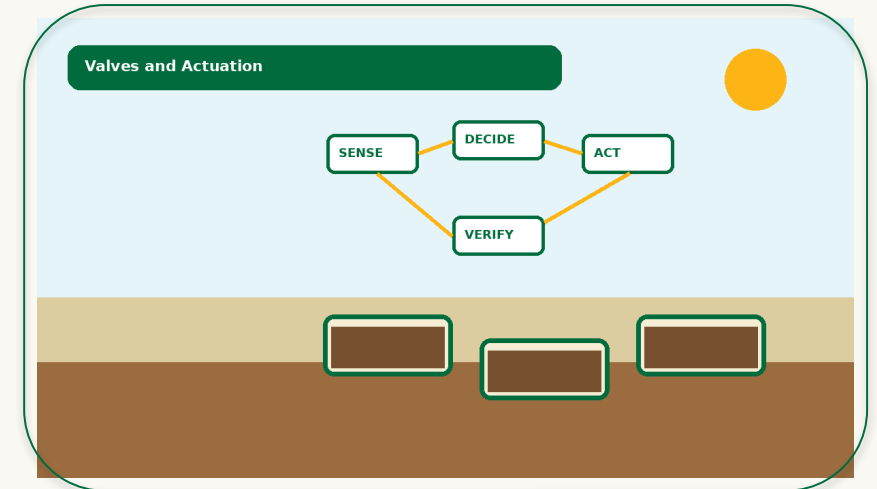
TEACHER PRACTICE

- Write a simple controller logic specification.

Valves and Actuation

Teacher technical knowledge, laboratory/field practice and student supervision

- Explain actuation as converting a controller command into a physical irrigation action.
- Identify normally open/closed behavior where relevant to installed valves.
- Verify valve state through safe observation and flow measurement.
- Protect wiring and connectors from water and mechanical damage.
- Adults handle pressurized disassembly and electrical repairs.



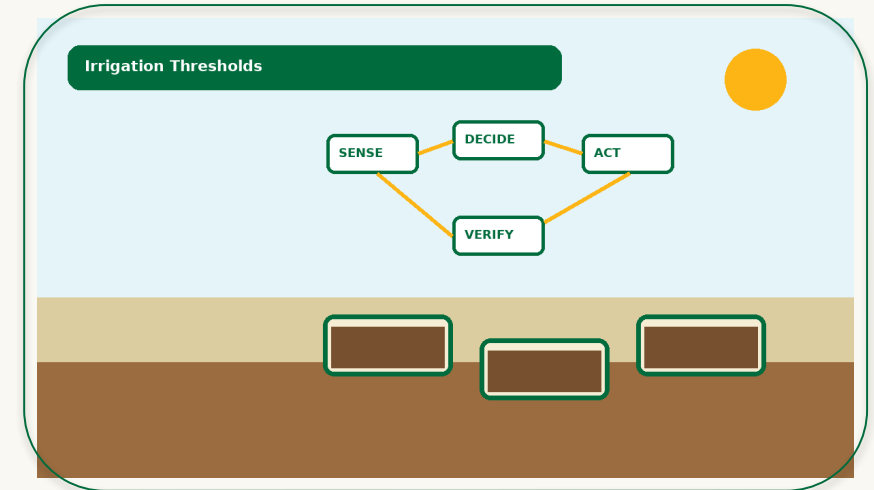
TEACHER PRACTICE

- Prepare a command-versus-actual-valve-state test.

Irrigation Thresholds

Teacher technical knowledge, laboratory/field practice and student supervision

- Define a threshold as a rule value that triggers or prevents irrigation.
- Base thresholds on crop/root-zone guidance, sensor characteristics and experimental purpose.
- Use upper/lower bounds or hysteresis conceptually to avoid rapid on-off cycling where appropriate.
- Document threshold changes.
- Do not use arbitrary values merely because they produce visible automation.



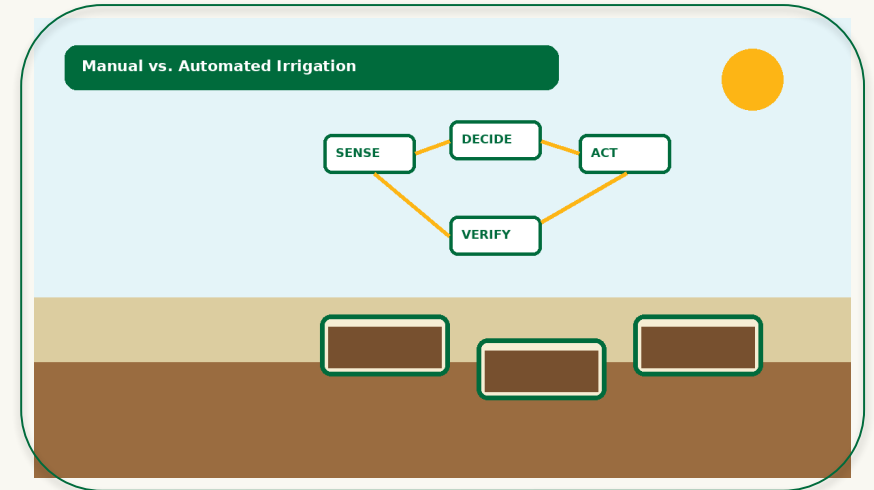
TEACHER PRACTICE

- Design and justify a simple moisture-threshold rule.

Manual vs. Automated Irrigation

Teacher technical knowledge, laboratory/field practice and student supervision

- Establish comparable manual and automated strategies with clearly defined rules.
- Measure water applied, soil-moisture behavior, crop response, labor and reliability.
- Keep crop, plot and infrastructure conditions as similar as feasible.
- Record interventions and system failures.
- Avoid assuming automation is more efficient before evaluation.



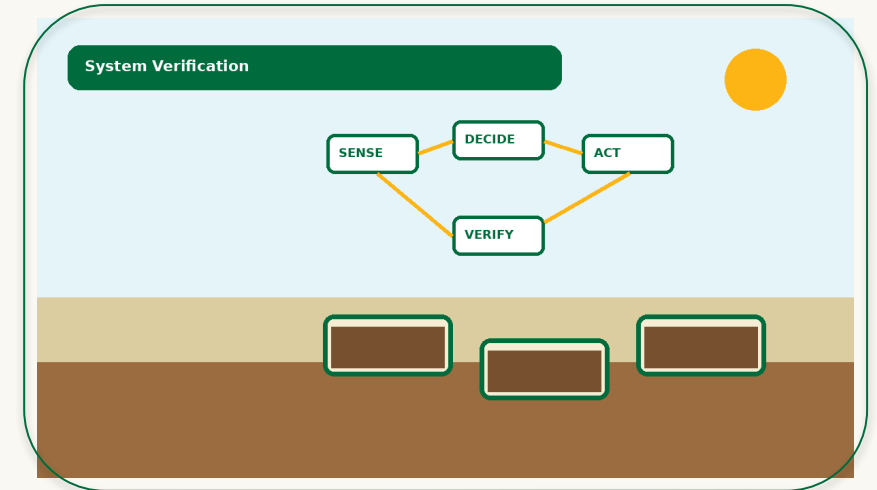
TEACHER PRACTICE

- Design a replicated manual-versus-automated comparison.

System Verification

Teacher technical knowledge, laboratory/field practice and student supervision

- Test sensor input, controller decision, valve actuation and measured water delivery as a complete chain.
- Use simulated inputs before live crop operation where appropriate.
- Test fail-safe and manual override behavior.
- Record expected versus observed outcomes.
- Correct and retest faults before routine use.



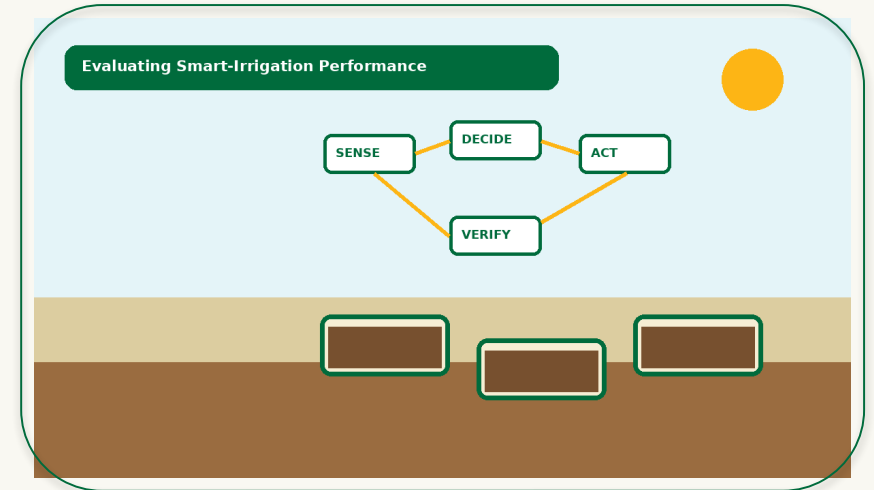
TEACHER PRACTICE

- Create a smart-irrigation verification checklist.

Evaluating Smart-Irrigation Performance

Teacher technical knowledge, laboratory/field practice and student supervision

- Evaluate water use, soil-moisture control, crop response, reliability, false/missed irrigation events and labor requirements.
- Use time-series and event records to assess controller behavior.
- Compare automated performance against the defined manual/reference strategy.
- Include sensor/actuator failures in evaluation.
- Recommend rule or hardware changes based on evidence.



TEACHER PRACTICE

- Completion task: prepare a smart-irrigation performance evaluation.